HANDLING AND MISSING DATA IN ETL ASSIGNMENT

SECTION A – THEORETICAL QUESTIONS

Q1. What are the most common reasons for missing data in ETL pipelines?

Answer

1. Uses error = form not fill properly.
2. System/software error= failed API calls / faulty sensors.
3. Optional field = users allow skip questions.
4. Data integration issues = format mismatch, encoding issues.
5. Human error.
6. Optional fields left blank by users.
7. Data corruption or loss during transfer.

Q2. Why is blindly deleting rows with missing values considered a bad practice in ETL?

Answer

- Leads to loss of valuable information
- Introduces bias if missingness is systematic (e.g., certain regions or customer types)
- Reduces dataset size, weakening statistical power

Q3. Explain the difference between: Listwise deletion Column deletion Also mention one scenario where each is appropriate.

Answer

Listwise deletion = delete entire row when any selected value missing.

Use only when missing % very small (5%).

Deletion column= delete column when column 60-70% value are missing.

Using for noncritical attributes.

Q4. Why is median imputation preferred over mean imputation for skewed data such as income?

Answer

Mean imputation = distorted variance

Fast, simple

Best for symmetric data

Median imputation = best for skewed data.

Does not affect outliers.

Common in pricing and outliers.

Mean is sensitive to extreme values (outliers), Median is robust and better represents central tendency in skewed distributions like income.

Q5. What is forward fill and in what type of dataset is it most useful?

Answer

Copies previous non-null value forward.

Great for continuous timelines.

Useful in **time-series datasets** (e.g., stock prices, sensor readings) where values change gradually.

Q6. Why should flagging missing values be done before imputation in an ETL workflow?

Answer

- Preserves information about missingness for later analysis.
- Helps detect patterns (e.g., certain groups consistently missing data).
- Missing values can carry meaning
- Helps predictive models
- Adds transparency

Q7. Consider a scenario where income is missing for many customers. How can this missingness itself provide business insights?

Answer

a. Customers hesitant to disclose income may belong to sensitive segments.
b. Could indicate privacy concerns or high-value customers unwilling to share details.
c. Missingness itself can guide targeted surveys or marketing strategies. Example = women do not want reveal their age.

Q 8,9,10 solved in google colab.